

Drug Formulation of Paracetamol Tablet

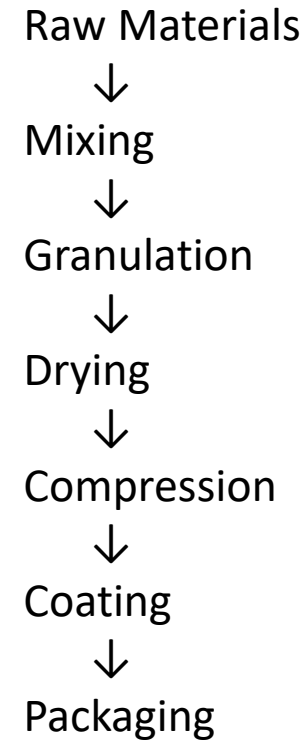
Introduction

Paracetamol is one of the most widely used pain-relieving and fever-reducing medicines. It belongs to the analgesic and antipyretic class. Available as tablets, capsules, syrups, and injections. Commonly used for headache, fever, and mild to moderate pain.



Formulation Process

- Weigh all ingredients accurately.
- Mix the active drug with excipients.
- Granulate the powder.
- Dry the granules.
- Compress into tablets using a tablet press.
- Coat the tablets (optional).
- Package and label the tablets.



Excipients Used

Excipient

Purpose

Microcrystalline Cellulose

Binder

Starch

Disintegrant

Magnesium Stearate

Lubricant

Talc

Glidant

Povidone

Binder

Lactose

Diluent

Stability Factors

- Protect from moisture.
- Store below 25°C.
- Avoid direct sunlight.
- Use moisture-proof packaging.
- Maintain proper humidity during storage.
- 🌡 Temperature
- 💧 Moisture
- ☀ Sunlight
- 📦 Packaging

Challenges

- Uniform mixing of ingredients. Prevent tablet breakage. Maintain drug stability. Ensure proper dissolution. Avoid contamination during manufacturing.



Conclusion

- Paracetamol tablets are safe and effective when manufactured correctly. Excipients improve tablet quality and performance. Proper formulation ensures stability, safety, and therapeutic effectiveness.



References

- World Health Organization (WHO)
- Indian Pharmacopoeia (IP)
- British Pharmacopoeia (BP)
- United States Pharmacopeia (USP)
- DrugBank
- FDA